Vortical Enclosure Dynamics, Vol. 3

From Closure Density to Gravity and Cosmology

Independent Research — 2026

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Abstract

Vortical Enclosure Dynamics (VED) derives physical structure from a single axiom: *there is difference.* Vol. 1 established the causal log C_{ij} and derived time and space. Vol. 2 showed that the Standard Model emerges from closure geometry.

This volume addresses gravity. The central claim is that gravity is not a new force, but a modification of the definition of distance. Incorporating closure density L_{ab} into the path cost yields, in a single coherent structure, the Einstein equation as the linear form of a fixed-point structure, the Schwarzschild solution as an exterior limit, the black hole interior as a non-equilibrium closure phase, and cosmic expansion as the global $\sigma > 0$ regime.

No additional structures are introduced beyond the closure framework. The present theory does not yet possess quantitative predictive power; structural organization is the primary goal.

Contents

1 Introduction

Vol. 1 of this series established that time and space emerge from the causal log C_{ij} through self-referential closure dynamics [1]. Vol. 2 showed that the gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, particle spectrum, and mass generation can be interpreted as structural consequences of triangular closure [2].

The present volume addresses the remaining question: how does gravitational dynamics arise within the same framework?

The starting point is the distance definition of Vol. 1:

$$d_{ij} = -\log \tilde{S}_{ij}, \qquad \tilde{S}_{ij} = \max_{\gamma:i\to j} \prod_{(ab)\in\gamma} S_{ab}. \tag{1}$$

This definition is purely structural, determined by the most permeable path in the causal log network. It yields a consistent geometry, but does not produce gravity. The reason is that it assumes structural homogeneity: no mechanism biases path selection according to accumulated closure structure.

Gravity, within VED, appears precisely as such a bias. The central claim of this volume is:

$$\boxed{\text{Gravity is not a new force. It appears as a modification of the definition of distance.}} \tag{2}$$

Incorporating closure density L_{ab} into the path cost unifies geometry and gravitational effect within a single selection principle. From this modification, the Einstein equation emerges as the linear approximation of a fixed-point structure, the Schwarzschild solution appears as an exterior limit, the black hole interior is identified as a non-equilibrium closure phase, and cosmic expansion arises as the global $\sigma > 0$ regime.

This interpretation is structural and qualitative in nature. No quantitative predictions are derived, and no claims are made about numerical correspondence with observed values. The goal is to exhibit a coherent geometric reading of gravitational physics.

The paper is organized as follows. Section 2 introduces the modified distance definition. Section 3 develops the closure tensor and its relation to the metric. Section 4 derives the Einstein equation as a fixed-point structure. Section 5 examines the exterior solution. Section 6 treats the black hole interior as a non-equilibrium phase. Section 7 addresses cosmic expansion. Section 8 synthesizes the three volumes. Section 9 discusses scope and limitations.

2 Where Does Gravity Come From?

2.1 The Distance of Vol.1 and Its Limitation

In Vol. 1, distance was defined purely from the causal log structure:

$$d_{ij} = -\log \tilde{S}_{ij}, \qquad \tilde{S}_{ij} = \max_{\gamma:i\to j} \prod_{(ab)\in\gamma} S_{ab}. \tag{3}$$

This definition selects the path of maximal propagation strength. Distance is therefore determined by the most permeable structural path.

$$\boxed{\text{Distance is determined solely by the structure of } C_{ij}.} \tag{4}$$

This construction yields a consistent geometry derived entirely from relational structure. However, an important limitation appears.

$$\boxed{\text{Gravity does not explicitly emerge from this definition.}} \tag{5}$$

The reason is that the distance above assumes structural homogeneity. All regions are treated equally except for variations encoded in propagation strength. No mechanism exists that biases path selection according to accumulated structure. Gravity, however, appears precisely as such a bias.

2.2 What Is Gravity?

In conventional physics, gravity is introduced either as a force acting between masses, or as curvature of spacetime. VED adopts a different starting point.

$$\boxed{\text{Gravity is the phenomenon by which closure density distorts the distance structure.}} \tag{6}$$

This represents a shift in perspective. The conventional approach introduces a new interaction. VED modifies the definition of distance itself.

$$\boxed{\text{Gravity is not added. It is a reinterpretation of existing structure.}} \tag{7}$$

Thus gravity is not an independent entity, but an emergent bias in the same structural geometry already defined in Vol. 1.

2.3 Extension of Distance

To incorporate this bias, assign a cost to each edge (a, b):

$$w_{ab} = -\log S_{ab} + \epsilon\, L_{ab}. \tag{8}$$

The first term represents geometric permeability, identical to Vol. 1. The second term introduces a congestion cost due to closure density. Distance is then defined by minimizing total path cost:

$$\boxed{d_{ij} = \min_{\gamma: i \to j} \sum_{(ab) \in \gamma} w_{ab}.} \tag{9}$$

This expression unifies geometry and gravitational effect within a single path selection principle. The overall structure is summarized in Figure 1.

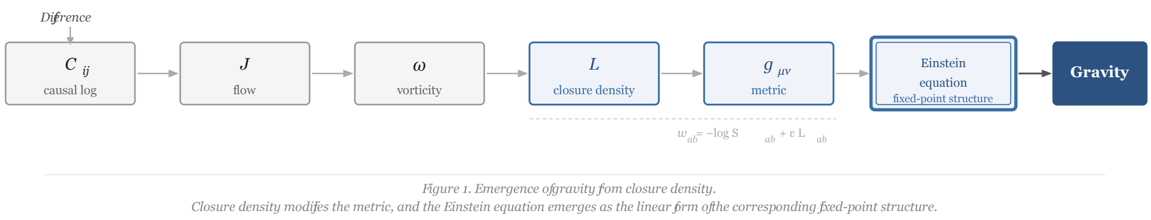

Figure 1: Emergence of gravity from closure density. Causal logs generate flow and vorticity, which produce closure density. Closure density modifies the metric, and the Einstein equation emerges as the linear form of the corresponding fixed-point structure. Gravity therefore emerges as a modification of distance rather than a fundamental force.

2.4 What the Unified Path Implies

It is essential that both terms in equation (8) are evaluated along the same path. If propagation strength and closure density were evaluated on different paths, the geometry would split into separate structures.

$$\boxed{\text{Gravity alters path selection itself.}} \tag{10}$$

This mirrors the conceptual shift introduced by general relativity. Einstein replaced force with curvature. VED replaces curvature with path cost. The two descriptions differ in interpretation but agree structurally.

2.5 Summary

By incorporating closure density into the distance definition, gravity emerges naturally from the same structural framework.

$$\boxed{\text{Gravity is not a new force.}} \tag{11}$$

$$\boxed{\text{It appears as a modification of the definition of distance.}} \tag{12}$$

3 Closure Density and Geometry

3.1 The Quantity Called Closure Density

Section 2 introduced L_{ab} as the additional contribution to the path cost:

$$\boxed{L_{ab} = \sum_k \phi_{abk}.} \tag{13}$$

This represents the total contribution of triangular closures containing the edge (a, b). Each triangle contributes to structural congestion, and the sum measures how densely that edge is embedded in closure structure. Intuitively, regions with many triangles are structurally dense; regions with few triangles are structurally sparse.

$$\boxed{\text{Closure density} = \text{density of structure.}} \tag{14}$$

This interpretation follows directly from Vol. 2, where triangular closure was identified as the minimal persistent structure.

3.2 Why It Distorts Distance

The modified edge cost was $w_{ab} = -\log S_{ab} + \epsilon L_{ab}$. The first term favors highly permeable paths; the second penalizes structurally congested regions. Thus, large closure density increases traversal cost.

$$\boxed{\text{Closure density effectively stretches distance.}} \tag{15}$$

This reproduces the qualitative behavior associated with gravity: dense regions become harder to traverse, propagation paths bend around them, and effective geometry is distorted. Unlike general relativity, however, curvature is not introduced as a fundamental geometric postulate. It emerges from the modification of path selection.

3.3 Tensorization

Closure density is directional. An edge contributes differently depending on orientation. To describe this, define a coarse-grained tensor over a region U:

$$L_{\mu\nu}(x) = \left\langle L_{ab}\, \hat{e}_\mu^{(ab)} \hat{e}_\nu^{(ab)} \right\rangle_U, \tag{16}$$

where $\hat{e}_\mu^{(ab)}$ is the direction vector determined internally from the distance structure. No external embedding space is assumed.

$$\boxed{L_{\mu\nu} = \text{closure tensor.}} \tag{17}$$

This tensor encodes directional congestion and determines anisotropic distance modifications.

3.4 Relation to the Metric

The modification of distance can be absorbed into an effective metric:

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + \alpha_G L_{\mu\nu}, \tag{18}$$

where $g_{\mu\nu}^{(0)}$ is the background metric derived in Vol. 1, $L_{\mu\nu}$ is the closure tensor, and α_G is a structural coupling constant.

$$\boxed{\text{The metric is determined by closure density.}} \tag{19}$$

This is structurally analogous to the statement that matter curves spacetime, but no matter field has been introduced. Geometry is determined entirely by closure structure.

3.5 G as an Effective Coupling

The coefficient α_G plays the role of gravitational coupling:

$$\boxed{G_{\text{eff}} \sim \alpha_G.} \tag{20}$$

This constant is not fundamental. It arises as an effective parameter of coarse-grained closure structure. At microscopic scales, different particle closures may contribute differently; at macroscopic scales, coarse-graining averages these contributions. This produces an approximately universal gravitational coupling.

$$\boxed{\text{The universality of } G \text{ emerges dynamically.}} \tag{21}$$

3.6 Summary

Closure density has been identified as the source of gravitational distortion.

$$\boxed{\text{Gravity} = \text{closure density.}} \tag{22}$$

The modification of distance induces an effective metric, and spacetime geometry emerges from coarse-grained closure structure.

$$\boxed{\text{Spacetime emerges as coarse-grained closure geometry.}} \tag{23}$$

4 The Einstein Equation as a Fixed-Point Structure

4.1 Positioning the Question

So far: distance is derived from C_{ij}; gravity appears as closure density L; the metric takes the form $g_{\mu\nu} = g^{(0)}_{\mu\nu} + \alpha_G L_{\mu\nu}$. The next question is therefore unavoidable:

$$\boxed{\text{What equation does this structure obey?}} \tag{24}$$

In conventional physics, the Einstein equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ is postulated as a fundamental law. Within VED, the situation is reversed.

$$\boxed{\text{The gravitational equation is not assumed.}} \tag{25}$$

Instead, it must emerge from the same self-consistent closure structure.

4.2 Starting Point: The Fixed-Point Equation

Vol. 1 introduced the self-referential structure:

$$C_{ij} = 1 - \exp(-\lambda \rho_i H_{ij}), \qquad \rho_i = \sum_j C_{ij}. \tag{26}$$

This defines a nonlinear fixed-point problem.

$$\boxed{\text{Structure determines itself.}} \tag{27}$$

Geometry, density, and flow are all determined simultaneously by this self-consistency condition. This self-referential nature is the key to the emergence of gravitational field equations.

4.3 Continuum Limit and Vorticity

Upon coarse-graining, define the flow $J_{ij} = C_{ij} - C_{ji}$. In the continuum limit:

$$\omega = \nabla \times J, \qquad L \sim |\omega|. \tag{28}$$

$$\boxed{\text{Closure density appears as vorticity of the flow.}} \tag{29}$$

This connects the microscopic closure structure to macroscopic geometric distortion.

4.4 Fixed-Point Condition

In the classical phase established in Vol. 1:

$$\sigma \approx 0, \qquad \nabla \cdot J \approx 0. \tag{30}$$

These conditions correspond to approximate conservation.

$$\boxed{\text{Conservation laws are properties of the phase.}} \tag{31}$$

Under these conditions, closure structure stabilizes and the metric becomes time-independent. This is the regime in which gravitational geometry emerges.

4.5 Self-Consistency Condition for the Metric

From Section 3, $g_{\mu\nu} = g^{(0)}_{\mu\nu} + \alpha_G L_{\mu\nu}$. However, $L_{\mu\nu}$ depends on flow, which depends on distance, which depends on the metric itself. Therefore $L_{\mu\nu} = L_{\mu\nu}[g]$, and the metric must satisfy:

$$\boxed{g_{\mu\nu} = g^{(0)}_{\mu\nu} + \alpha_G L_{\mu\nu}[g].} \tag{32}$$

This defines a fixed-point problem for the metric.

4.6 Linearization and the Form of the Equation

In the weak-gravity limit $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ ($|h_{\mu\nu}| \ll 1$), expanding equation (32) to first order yields:

$$\boxed{G_{\mu\nu} = 8\pi G_{\text{eff}} L_{\mu\nu}.} \tag{33}$$

$$\boxed{\text{Structurally analogous to the Einstein equation.}} \tag{34}$$

Thus the Einstein equation appears as the linear approximation of the self-consistent closure structure.

4.7 The Inversion

Conventional understanding: the Einstein equation = a fundamental law.

VED:

$$\boxed{\text{The Einstein equation is a linear approximation of a fixed-point structure.}} \tag{35}$$

It holds in the classical phase where closure is stable, but need not hold outside this regime. Gravity is therefore not fundamental but emergent.

4.8 Why General Relativity Works

This interpretation explains the empirical success of GR.

$$\boxed{\text{GR is not incorrect.}} \tag{36}$$

$$\boxed{\text{It is an effective theory of the classical closure phase.}} \tag{37}$$

In this phase, closure density varies slowly, conservation approximately holds, and the metric is stable. Under these conditions, the Einstein equation provides an accurate description.

5 The Schwarzschild Solution as Exterior Closure

5.1 What This Section Does

Section 4 showed that gravitational dynamics arise as a fixed-point structure of closure density. The next step is to examine the simplest configuration: static, spherically symmetric, and with localized closure density.

$$\boxed{\text{Closure concentrated at a single point.}} \tag{38}$$

In general relativity, this leads to the Schwarzschild solution. Within VED, the same structure emerges from modification of distance.

5.2 Definition of the Exterior Vacuum

Consider a localized closure distribution $L(x) = M\,\delta(x)$. Outside the localized region $(r > R)$, $L(r) = 0$. In this region, closure density vanishes but the metric remains influenced by the boundary conditions imposed by the central structure.

The potential equation becomes:

$$\nabla^2\Psi \propto L = 0 \qquad \Rightarrow \qquad \nabla^2\Psi = 0. \tag{39}$$

For spherical symmetry, the solution is:

$$\boxed{\Psi(r) = -\frac{G_{\text{eff}}M}{r}.} \tag{40}$$

$$\boxed{\text{The Newtonian potential.}} \tag{41}$$

5.3 Form of the Metric

In the weak-gravity limit, the metric takes the form:

$$\boxed{ds^2 \approx -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 + \frac{2GM}{r}\right) dr^2 + r^2 d\Omega^2.} \tag{42}$$

This expression corresponds to the first-order expansion of the Schwarzschild metric.

$$\boxed{\text{Consistent with the Schwarzschild solution (weak-gravity limit).}} \tag{43}$$

The full nonlinear expression emerges when the fixed-point equation is solved beyond linear order.

5.4 What Is Happening

General relativity interprets this solution as curvature of spacetime. VED interprets it differently.

$$\boxed{\text{Distance cost is modified by closure density.}} \tag{44}$$

As a result, propagation paths change, light trajectories bend, time dilation appears, and an effective metric emerges. The observable consequences coincide, while the interpretation differs.

5.5 VED Interpretation of the Horizon

The Schwarzschild radius is $r_s = 2GM$. At this radius, $g_{tt} \to 0$. Within VED, this does not represent geometric breakdown.

$$\boxed{\text{The horizon is a critical point of path structure.}} \tag{45}$$

At this point, path cost diverges, propagation freezes, and closure density approaches a critical value. The horizon is therefore a phase boundary, as illustrated in Figure 2.

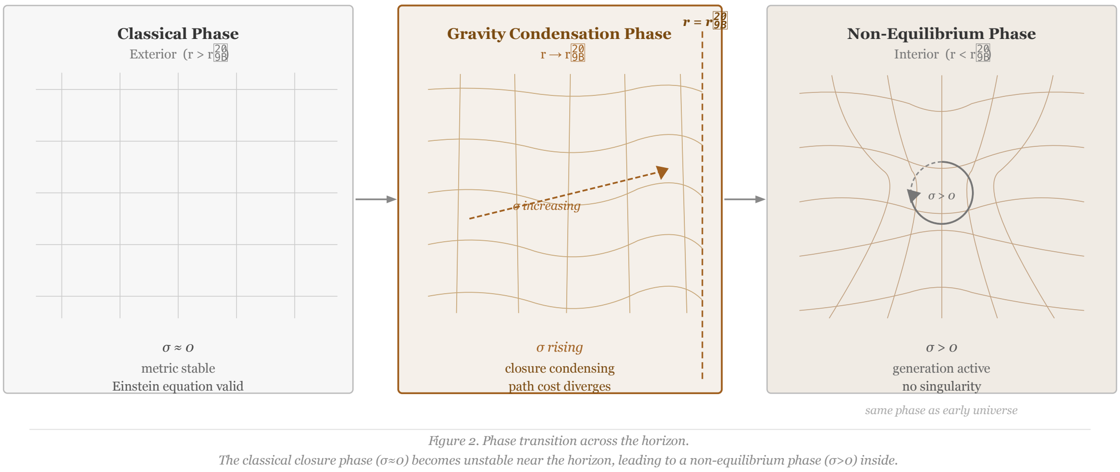

Figure 2: Phase transition across the horizon. The classical closure phase ($\sigma \approx 0$) becomes unstable near the horizon (gravity condensation phase), leading to a non-equilibrium phase ($\sigma > 0$) inside. The horizon corresponds to the transition region.

5.6 Summary

The exterior gravitational solution emerges naturally from closure density.

$$\boxed{\text{The Schwarzschild solution appears as a limit of closure geometry.}} \tag{46}$$

No new force or curvature postulate is required.

$$\boxed{\text{Gravity emerges from modified distance.}} \tag{47}$$

6 Black Hole Interior as a Non-Equilibrium Phase

6.1 Revisiting the Meaning of the Horizon

Section 5 showed that the Schwarzschild radius $r_s = 2GM$ appears as a critical point of the modified distance structure. At this radius, propagation cost diverges and the exterior solution ceases to apply.

$$\boxed{\text{The horizon is a phase boundary.}} \tag{48}$$

This interpretation differs from the conventional geometric breakdown. Instead, the horizon marks a transition between two regimes of closure dynamics.

6.2 Exterior vs. Interior

Outside the horizon, the classical phase holds:

$$\sigma \approx 0, \qquad \nabla \cdot J \approx 0, \qquad \partial_t C \approx 0. \tag{49}$$

In this regime, closure structure is stable, conservation laws hold, and the metric is well-defined.

$$\boxed{\text{Exterior} = \text{classical closure phase.}} \tag{50}$$

Crossing the horizon changes these conditions.

6.3 Reignition of σ

Inside the horizon, the generation term becomes nonzero:

$$\boxed{\sigma > 0.} \tag{51}$$

Log generation resumes and the fixed-point structure breaks down: $\partial_t C \neq 0$. This implies that geometry is no longer static.

$$\boxed{\text{Time begins to be generated again.}} \tag{52}$$

The interior therefore corresponds to a non-equilibrium regime of closure.

6.4 Collapse of the Fixed Point

In the exterior, $\partial_t C \approx 0$; in the interior, $\partial_t C \neq 0$. The metric is no longer determined by a fixed-point condition.

$$\boxed{\text{The fixed-point structure collapses.}} \tag{53}$$

Geometry becomes dynamically generated rather than static.

6.5 The Singularity Does Not Exist

General relativity predicts a singularity at the center. VED prohibits perfect closure. From Vol. 1:

$$B(L) = -\kappa_B \log\left(1 - \frac{L}{L_{\max}}\right) \to \infty \qquad (L \to L_{\max}). \tag{54}$$

$$\boxed{\text{Perfect closure is dynamically forbidden.}} \tag{55}$$

Therefore, the center is not a singularity but an unreachable limit.

6.6 Interior Dynamics

With $\sigma > 0$, closure structure is continuously regenerated.

$$\boxed{\text{The interior is a region of structural regeneration.}} \tag{56}$$

Logs are generated, closure reorganizes, and geometry evolves. The interior is therefore dynamically active rather than a state of collapse.

6.7 The Inversion

Conventional interpretation: black hole = endpoint.

VED:

$$\boxed{\text{Black hole} = \text{entry into a new phase.}} \tag{57}$$

The exterior is a fixed world; the interior is a world under generation.

6.8 Correspondence with the Early Universe

The early universe is characterized by $\sigma > 0$. The black hole interior also satisfies $\sigma > 0$.

$$\boxed{\text{The early universe and black hole interior are the same phase at different scales.}} \tag{58}$$

6.9 Conclusion

$$\boxed{\text{The black hole interior is not a singularity.}} \tag{59}$$

$$\boxed{\text{It is a non-equilibrium closure phase.}} \tag{60}$$

Closure continues to evolve, and geometry remains dynamically generated.

7 Cosmology: The Global $\sigma > 0$ Phase

7.1 Shift in the Question

Section 6 identified the black hole interior as a local non-equilibrium closure phase characterized by $\sigma > 0$. This raises the natural global question:

$$\boxed{\text{What happens when the } \sigma > 0 \text{ phase extends across the entire universe?}} \tag{61}$$

Within the VED framework, the answer is direct:

$$\boxed{\text{Cosmic expansion emerges as a global } \sigma > 0 \text{ phase.}} \tag{62}$$

Thus cosmology is interpreted not as an initial condition, but as a phase of the closure dynamics.

7.2 The Fundamental Equation

From Vol. 1, the continuity equation for log density is $\partial_t \rho + \nabla \cdot J = \sigma$. Under the homogeneous and isotropic approximation, $\nabla \cdot J \approx 3H\rho$, which yields:

$$\boxed{\dot{\rho} + 3H\rho = \sigma.} \tag{63}$$

The first term represents dilution due to expansion; the second represents generation of structure.

7.3 The Equation in τ-Time

Using the time definition from Vol. 1, $d\tau/dt = \rho$, the equation becomes:

$$\boxed{\frac{d\rho}{d\tau} + 3H_\tau \rho = \sigma(\rho).} \tag{64}$$

The scale factor is defined structurally as $a(\tau) \sim \langle d_{ij}(\tau) \rangle$. Since the expansion rate is determined by density, $H_\tau \sim \rho$, giving:

$$H^2 \propto \rho, \tag{65}$$

which is structurally analogous to the Friedmann equation.

7.4 Final Form

Substituting $H_\tau \sim \rho$, the evolution equation becomes:

$$\boxed{\frac{d\rho}{d\tau} + 3\gamma\rho^2 = \sigma(\rho).} \tag{66}$$

Here $3\gamma\rho^2$ represents dilution due to expansion and $\sigma(\rho)$ represents generation of logs. Cosmology therefore emerges as competition between generation and dilution.

7.5 The Form of σ

From the self-referential structure of Vol. 1, the minimal effective form is:

$$\boxed{\sigma(\rho) = \xi\rho\left(1 - \frac{\rho}{\rho_{\max}}\right).} \tag{67}$$

This corresponds to growth at low density and saturation at high density. The logistic form arises naturally from self-referential selection.

7.6 Fixed Point and de Sitter Phase

The fixed-point solution satisfies $3\gamma\rho_*^2 = \sigma(\rho_*)$, giving:

$$\rho_* = \frac{\xi}{\xi/\rho_{\max} + 3\gamma}. \tag{68}$$

At the fixed point, $H_* = \gamma\rho_* = \text{const}$, and therefore:

$$\boxed{a(\tau) \sim e^{H_*\tau}.} \tag{69}$$

$$\boxed{\text{de Sitter-type accelerated expansion.}} \tag{70}$$

7.7 Origin of the Cosmological Constant

The effective cosmological constant is:

$$\boxed{\Lambda_{\text{eff}} \sim \gamma^2\rho_*^2.} \tag{71}$$

The cosmological constant is therefore not fundamental, but determined by closure structure.

7.8 The Inversion

Conventional cosmology: expansion = initial condition.

VED:

$$\boxed{\text{Cosmic expansion} = \text{phase selection.}} \tag{72}$$

If $\sigma = 0$, the universe is static. If $\sigma > 0$, the universe expands.

7.9 Unification with Black Holes

The black hole interior satisfies $\sigma > 0$; the expanding universe also satisfies $\sigma > 0$.

$$\boxed{\text{Black hole interior and cosmic expansion are the same phase at different scales.}} \tag{73}$$

7.10 Conclusion

Cosmology emerges as global closure dynamics.

$$\boxed{\text{The universe is continuously generated.}} \tag{74}$$

$$\boxed{\text{Expansion is a consequence of } \sigma > 0.} \tag{75}$$

The large-scale dynamics of the universe follow directly from closure structure.

8 Synthesis: The World as Structure

8.1 The Flow of This Theory

This work began from a single axiom:

$$\boxed{\text{There is difference.}} \tag{76}$$

From difference, gradients emerge. From gradients, flow appears. From flow, vorticity forms. From vorticity, closure develops.

$$\boxed{\text{Structure emerges from dynamics.}} \tag{77}$$

8.2 The Overall Structure

The entire theory can be summarized by a single chain:

$$\boxed{\text{Difference} \to C_{ij} \to J \to \omega \to L \to g \to \text{Einstein equation} \to \text{Universe}} \tag{78}$$

Quantity	Meaning
C_{ij}	causal log
J	flow of logs
ω	vorticity
L	closure density
g	effective metric
Einstein equation	fixed-point structure
Universe	global $\sigma > 0$ phase

$$\boxed{\text{Everything forms a single structural chain.}} \tag{79}$$

8.3 Correspondence Across Volumes

The three volumes form a unified progression.

Vol. 1 — Generation: causal log C_{ij}; time $d\tau/dt = \rho$; space $d_{ij} = -\log \tilde{S}_{ij}$. Structure is generated.

Vol. 2 — Structure: triangular closure; gauge symmetry $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$; particles, charge, mass. Structure stabilizes.

Vol. 3 — Dynamics: closure density $\to$ gravity; fixed-point structure $\to$ Einstein equation; $\sigma > 0$ phase $\to$ cosmic expansion. Structure evolves.

$$\boxed{\text{Generation, structure, and dynamics are unified.}} \tag{80}$$

8.4 Reinterpreting Conservation Laws

Conventional physics treats conservation laws as fundamental; violations appear as exceptions. Within VED, the continuity equation is:

$$\partial_t \rho + \nabla \cdot J = \sigma. \tag{81}$$

Two regimes appear: $\sigma = 0$ implies conservation holds; $\sigma > 0$ implies generation occurs.

$$\boxed{\text{Conservation laws are properties of a phase.}} \tag{82}$$

Conservation is therefore not fundamental, but emergent.

8.5 Reinterpreting Gravity

Conventional interpretations treat gravity as force or as curvature. VED interprets it differently.

$$\boxed{\text{Gravity} = \text{modification of distance.}} \tag{83}$$

$$\boxed{\text{Gravity} = \text{closure density.}} \tag{84}$$

8.6 Black Holes and the Universe

The black hole interior satisfies $\sigma > 0$; the expanding universe also satisfies $\sigma > 0$.

$$\boxed{\text{Black hole interior and cosmic expansion are the same phase.}} \tag{85}$$

8.7 The Final Inversion

Conventional physics begins by assuming spacetime, fields, and symmetry. VED instead derives them from structure.

$$\boxed{\text{Everything emerges from closure structure.}} \tag{86}$$

Time is generated. Space is relational. Particles are closure attractors. Forces are flow. The universe is a phase.

8.8 The Final Statement

$$\boxed{\text{The world is not closed.}} \tag{87}$$

$$\boxed{\text{It exists by never fully closing.}} \tag{88}$$

This statement serves both as the starting point and the conclusion of VED.

$$\text{Difference} \rightarrow C_{ij} \rightarrow J \rightarrow \omega \rightarrow L \rightarrow g_{\mu\nu} \rightarrow \text{Einstein equation} \rightarrow \text{Universe} \tag{89}$$

$$\boxed{\begin{aligned} \text{Time} &= \text{quantity of logs} \\ \text{Space} &= \text{structure of logs} \\ \text{Particle} &= \text{closure attractor} \\ \text{Gravity} &= \text{potential from closure density} \\ \text{Universe} &= \text{phase of continuous generation} \end{aligned}} \tag{90}$$

$$\boxed{\text{Everything emerges from a single root: } C_{ij}.} \tag{91}$$

9 Discussion

9.1 Relation to Vol. 1 and Vol. 2

Vol. 1 established time and space as emergent from the causal log C_{ij}. Vol. 2 showed that particle physics structure emerges from triangular closure. The present volume completes this picture by deriving gravitational dynamics from closure density.

The key unifying element is the closure energy functional. In Vol. 1, it governed the emergence of metric structure. In Vol. 2, its local minima were identified with particle types. In Vol. 3, its coarse-grained form determines the gravitational field. All three volumes therefore share a common geometric foundation.

9.2 Scope and Limitations

The interpretations advanced in this volume are structural and qualitative. Several points of caution are warranted.

First, the identification of $G_{\mu\nu} = 8\pi G_{\text{eff}} L_{\mu\nu}$ with the Einstein equation is structural. No derivation is given of the precise numerical coefficient 8π, nor of the relation between G_{eff} and observed gravitational coupling. A rigorous derivation would require a more developed formalism.

Second, the Schwarzschild metric is recovered only in the weak-gravity limit. The full nonlinear solution is asserted to emerge from the exact fixed-point equation, but this has not been demonstrated explicitly.

Third, the cosmological equation (66) is structurally analogous to the Friedmann equation, but the precise correspondence between ρ and energy density, and between γ and the equation of state, has not been established.

Fourth, the claim that the black hole interior corresponds to a $\sigma > 0$ phase, while motivated by the prohibition of perfect closure, remains a qualitative assertion. A quantitative treatment would require extending the fixed-point formalism into the non-equilibrium regime.

9.3 Relation to General Relativity

The present framework does not compete with general relativity. GR is accurate in the classical closure phase ($\sigma \approx 0$), and the present work is consistent with this. The distinction arises in other phases: at the black hole interior, at the early universe, and at any regime where the fixed-point structure breaks down.

9.4 Path Forward

Several directions suggest themselves for future development.

The most immediate is a more rigorous treatment of the continuum limit: deriving the closure tensor $L_{\mu\nu}$ explicitly from the discrete network, and showing that the linearized fixed-point equation takes the form $G_{\mu\nu} = 8\pi G_{\text{eff}} L_{\mu\nu}$ without additional assumptions.

A second direction is the cosmological model. The logistic form of $\sigma(\rho)$ is motivated by self-referential selection, but its precise form should be derivable from the fixed-point structure. Connecting the fixed-point density ρ_* to observed cosmological parameters would constitute a non-trivial test of the framework.

A third direction concerns the non-equilibrium phase. The interior dynamics of $\sigma > 0$ regions — including the black hole interior and the early universe — require a treatment beyond the classical fixed-point approximation. This may connect to information-theoretic questions about black hole evaporation and the origin of entropy.

10 Conclusion

This volume has shown that gravitational dynamics can be interpreted as arising from closure geometry. Beginning from the modified distance definition $w_{ab} = -\log S_{ab} + \epsilon L_{ab}$, we have identified:

- gravity as a modification of distance, not a new force (Section 2);
- the closure tensor $L_{\mu\nu}$ as the source of metric distortion (Section 3);
- the Einstein equation as the linear approximation of a fixed-point structure (Section 4);
- the Schwarzschild solution as the exterior limit of closure geometry (Section 5);
- the black hole interior as a non-equilibrium closure phase with $\sigma > 0$ (Section 6);
- cosmic expansion as the global $\sigma > 0$ regime, structurally analogous to the Friedmann equation (Section 7).

The overall logical chain is:

$$\text{Difference} \rightarrow C_{ij} \rightarrow J \rightarrow \omega \rightarrow L \rightarrow g_{\mu\nu} \rightarrow \text{Einstein equation} \rightarrow \text{Universe.} \tag{92}$$

Taken together with Vol. 1 and Vol. 2, this completes the structural derivation of time, space, matter, and gravity from a single axiom: *there is difference.*

The interpretation is structural and qualitative in nature, and does not yet constitute a predictive theory. Quantitative correspondence with observation remains a goal for future work.

$$\boxed{\text{The world is not closed. It exists by never fully closing.}} \tag{93}$$

References

[1] Yuhki Endoh. Vortical enclosure dynamics, vol. 1: From difference to space and time, 2026. Independent research draft.

[2] Yuhki Endoh. Vortical enclosure dynamics, vol. 2: From closure geometry to the standard model, 2026. Independent research draft.

A Notation

Symbol	Definition	Source
C_{ij}	causal log matrix	Vol. 1
$\rho_i = \sum_j C_{ij}$	log density	Vol. 1
S_{ab}	symmetric component of C_{ij}	Vol. 1
$\tilde{S}_{ij}$	maximal path propagation strength	Vol. 1
d_{ij}	weighted shortest distance	Vol. 3 §2
$w_{ab} = -\log S_{ab} + \epsilon L_{ab}$	modified edge cost	Vol. 3 §2
$L_{ab} = \sum_k \phi_{abk}$	local closure density	Vol. 2+Vol. 3
$L_{\mu\nu}$	closure tensor (coarse-grained)	Vol. 3 §3
$J_{ij} = C_{ij} - C_{ji}$	log flow (antisymmetric)	Vol. 3
$\omega = \nabla \times J$	vorticity	Vol. 3
σ	log generation rate (source of time)	Vol. 3
α_G	gravitational coupling constant	Vol. 3
$G_{\text{eff}} \sim \alpha_G$	effective gravitational constant	Vol. 3 §3
$g^{(0)}_{\mu\nu}$	background metric from Vol. 1	Vol. 1
$g_{\mu\nu} = g^{(0)}_{\mu\nu} + \alpha_G L_{\mu\nu}$	effective metric	Vol. 3 §3
Ψ	gravitational potential	Vol. 3 §4
$r_s = 2G_{\text{eff}}M$	Schwarzschild radius	Vol. 3 §4
$\sigma(\rho)$	effective model of log generation rate	Vol. 3 §6
ρ_*	cosmological fixed-point density	Vol. 3 §6
H_*	de Sitter expansion rate	Vol. 3 §6
$\Lambda_{\text{eff}} \sim \gamma^2 \rho_*^2$	effective cosmological constant	Vol. 3 §6